Name: _____ Class and student number: _____

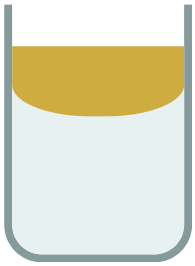
Carry out the investigation and collect data

Change one factor at a time. First set up a group without heating to use for comparison.

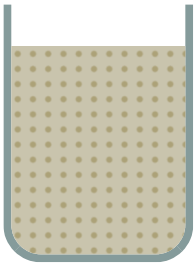
✦Fair test: oil, water, temperature and time are fixed. Only the combination of digestive juices changes.

Observe the static images, choose and record the appearances. The images show the preset results of this VL.

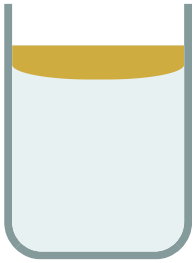
Control group: add 2 mL of water



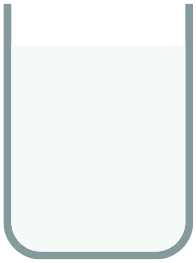
Add 2 mL of digestive juice X



Add 2 mL of digestive juice Y



Add 1 mL each of digestive juices X and Y



Name: _____ Class and student number: _____

Select the appearance based on what you observe:

Experiment setup	First confirmed answer • Your recorded appearance
Control group: add 2 mL of water	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 2 mL of digestive juice X	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 2 mL of digestive juice Y	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear
Add 1 mL each of digestive juices X and Y	☐ Separate oil and water layers (no clear change you can see) ☐ Separate oil and water layers (thinner oil layer) ☐ become cloudy ☐ become clear

If you revise an answer, keep the first answer and write the change here.

Name: _____ Class and student number: _____

03 • Data and explanations

Complete the conclusions using your data

1. What does digestive juice X do?

Digestive juice X mainly causes lipids to form _____.

Choose one.

☐ a clear solution ☐ a cloudy emulsion (乳狀液)

2. How does digestive juice X help digestive juice Y act?

Digestive juice X first breaks up large oil droplets into small droplets, causing their total surface area to _____, which helps digestive juice Y act.

Choose one.

☐ decrease ☐ increase

3. What can this experiment tell us?

We only observed the oil layer and whether the mixture was cloudy or clear. We did not test for digestion products. Which statement fits best?_____

Choose one.

☐ Appearance provides clues about the effects of digestive juices, but alone cannot prove that glycerol (甘油) and fatty acids (脂肪酸) have formed.

☐ A clear mixture completely proves that lipids have broken down into glycerol (甘油) and fatty acids (脂肪酸).

If you revise an answer, keep the first answer and write the change here.

Name: _____ Class and student number: _____

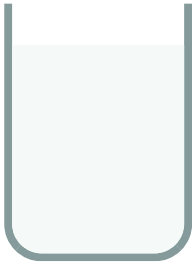
Extension: the effects of heating on digestive juices

Which digestive juice is affected by high temperature?

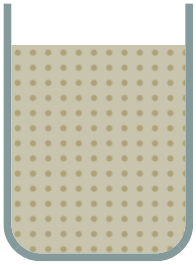
Use “add 1 mL each of digestive juices X and Y” as the group for comparison. Boil only one digestive juice at a time. Complete both tests and record your observations.

Cool each boiled digestive juice to the same reaction temperature of 37°C before comparing.

Add 1 mL each of digestive juices X and Y (boil X, then cool)



Add 1 mL each of digestive juices X and Y (boil Y, then cool)



Select the appearance based on what you observe:

Experiment setup	First confirmed answer • Your recorded appearance
Add 1 mL each of digestive juices X and Y (boil X, then cool)	☐ become cloudy ☐ become clear
Add 1 mL each of digestive juices X and Y (boil Y, then cool)	☐ become cloudy ☐ become clear

Extension conclusion: the digestive juice that loses its function after boiling is_____.

Choose one.

☐ Digestive juice X ☐ Digestive juice Y

If you revise an answer, keep the first answer and write the change here.